Overview of Principal Component Analysis (PCA)

PCA for Process and Quality Monitoring

Unit III Lecture Notes

Mathematics for Data Science

Learning objectives

By the end of this lecture, you should be able to:

- Explain the purpose and main idea of PCA.
- Describe eigenvalues, eigenvectors, principal components, and explained variance.
- Outline the steps used to carry out PCA.
- Explain how PCA can be used to monitor multivariate process quality.
- Interpret score monitoring, Hotelling's T^2, and residual Q statistics at an introductory level.

What is Principal Component Analysis?

Definition

Principal Component Analysis (PCA) is a statistical method that transforms a set of possibly correlated variables into a smaller set of uncorrelated variables called **principal components**.

- The first principal component (PC1) captures the greatest possible variance.
- PC2 captures the greatest remaining variance and is orthogonal to PC1.
- Later components explain progressively less variance.

Why use PCA? To reduce dimensionality, summarize patterns, visualize data, reduce noise, and support process monitoring.

Geometric intuition

Imagine measurements plotted in two dimensions.

- PCA finds a direction along which the data vary most.
- The second direction is perpendicular to the first and captures the next-largest variation.
- Projecting observations onto these directions gives their **component scores**.

Important

PCA summarizes variation. A high-variance direction is not automatically the most useful direction for predicting a target variable.

Mathematical foundation

For a random vector X with mean vector μ and covariance matrix

$$\Sigma = E[(X - \mu)(X - \mu)^T],$$

the variance of a linear combination $a^T X$ is

$$\mathrm{Var}(a^T X) = a^T \Sigma a.$$

The first component solves

$$\max_a a^T \Sigma a \quad \text{subject to } a^T a = 1.$$

Its direction is an eigenvector of Σ associated with the largest eigenvalue.

Eigenvalues and eigenvectors

- **Eigenvector:** direction of a principal component.
- **Eigenvalue:** variance captured by that component.
- Sort eigenvalues from largest to smallest: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$.

The proportion of total variance explained by PC i is

$$\frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j} \times 100\%.$$

The cumulative proportion for the first k components is

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{j=1}^{p} \lambda_j} \times 100\%.$$

PCA procedure

1. Arrange the data matrix X (rows = observations, columns = variables).
2. Center each variable by subtracting its sample mean.
3. Standardize variables if their scales differ and correlation-based PCA is appropriate.
4. Calculate the covariance matrix S or correlation matrix.
5. Find eigenvalues and eigenvectors.
6. Order components by decreasing eigenvalue.
7. Calculate explained and cumulative variance percentages.
8. Retain the number of components justified by the problem or study.
9. Calculate scores: $Z = X_c V_k$.

Worked example: explained variance

Suppose the eigenvalues are

$$\lambda_1 = 6, \qquad \lambda_2 = 3, \qquad \lambda_3 = 1.$$

Total variance:

$$6 + 3 + 1 = 10.$$

Component	Calculation	Variance explained
PC1	$6/10 \times 100$	60%
PC2	$3/10 \times 100$	30%
PC3	$1/10 \times 100$	10%
PC1 + PC2	$(6+3)/10 \times 100$	90%

If the requirement is to retain at least 90% of the variance, retain the first two components.

What is quality monitoring?

Goal

Quality monitoring checks whether a process continues to behave like its established, stable operating condition.

In modern processes, many variables are measured together, for example:

- temperature, pressure, flow rate, and vibration;
- product dimensions, weight, strength, and finish;
- chemical concentrations or sensor readings.

Variables can be correlated. Monitoring each variable separately may miss unusual combinations of measurements. PCA compresses the measurements into a smaller number of scores while retaining important variation.

Why use PCA for quality monitoring?

- Handles many correlated process variables.
- Summarizes normal process variation using a small number of components.
- Helps identify unusual combinations of measurements.
- Supports visual monitoring with score plots.
- Can detect changes in the modeled variation and in residual variation.

Caution

PCA is not a substitute for process knowledge. Data preparation, stable baseline data, and appropriate control limits are essential.

PCA-based monitoring: training stage

1. Collect historical data from a process believed to be stable.
2. Check missing values, sensor errors, and obvious data-quality problems.
3. Center variables; standardize if needed.
4. Fit PCA and choose retained components using a documented criterion.
5. Calculate the baseline score and residual distributions.
6. Establish suitable control limits for the chosen monitoring statistics.

The baseline should represent normal operation. If the training data contain faults or multiple operating regimes, the model and limits may be misleading.

PCA-based monitoring: ongoing stage

For each new observation:

1. Apply the same centering and scaling used for the baseline.
2. Project the observation into the retained PCA space.
3. Calculate monitoring statistics.
4. Compare the statistics with their established control limits.
5. Investigate signals using process knowledge and follow-up checks.

A signal indicates an unusual observation relative to the baseline model; it does not by itself prove a specific cause or defect.

Two common monitoring statistics

1. Hotelling's T^2 statistic

$$T^2 = \sum_{i=1}^{k} \frac{t_i^2}{\lambda_i},$$

where t_i is the score on retained component i and λ_i is its modeled variance. It measures how extreme a point is within the retained PCA model.

2. Residual statistic, often called Q

$$Q = \|x_c - \hat{x}_c\|^2,$$

where $\hat{x}_c$ is the reconstruction using retained components. It measures variation not represented by the retained PCA model.

Interpreting monitoring signals

Signal	Possible interpretation	Follow-up
High T^2	Unusual combination or magnitude within modeled component space	Check scores, variables, operating conditions, and sensor records
High Q	New pattern not well represented by retained components	Inspect residuals and variables with large reconstruction errors
Both high	Strong evidence of a departure from the baseline model	Investigate promptly and confirm using process evidence

Control limits depend on the method, assumptions, sample size, and process. Do not use arbitrary universal limits.

Mini worked example: score monitoring

Suppose a two-component model has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = 2$, and a new observation has scores $t_1 = 2$ and $t_2 = 1$.

$$T^2 = \frac{2^2}{4} + \frac{1^2}{2} = 1 + 0.5 = 1.5.$$

This gives a monitoring statistic of 1.5.

- Compare 1.5 with the control limit established for the fitted process model.
- If it exceeds the limit, flag the observation for investigation.
- The value alone cannot determine whether the product is defective.

Strengths and limitations

Strengths

- Reduces many variables to a few interpretable summaries.
- Can detect multivariate patterns missed by separate univariate charts.
- Provides both score-space and residual-space monitoring.

Limitations

- Components may be difficult to interpret physically.
- Scaling and baseline selection affect results.
- Nonlinear patterns or changing operating modes may need other methods.
- A signal requires investigation; PCA alone does not diagnose the cause.

Exam-focused questions

1. Define PCA and state three applications.
2. Explain the roles of eigenvalues and eigenvectors in PCA.
3. Given eigenvalues, calculate individual and cumulative variance explained.
4. List the steps for carrying out PCA.
5. Explain why PCA is useful in multivariate quality monitoring.
6. Distinguish Hotelling's T^2 and the residual Q statistic.
7. Describe the training and ongoing monitoring stages.
8. State two advantages and two limitations of PCA-based monitoring.

Summary

- PCA converts correlated variables into ordered, uncorrelated components.
- Eigenvectors define component directions; eigenvalues quantify component variance.
- Explained-variance percentages guide dimensionality reduction.
- Quality monitoring uses PCA scores and residuals to detect departures from a stable baseline.
- Control limits must be established appropriately, and signals must be investigated.

Questions?